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4 / 4 submitted

Q1. Arrays, Array Applications and Structures

CODE

```
#include<stdio.h>
int main(){
    char str[100];
    int i,j,count;
    scanf("%s", str);
    for(i=0;str[i]!='\0';i++){
        count=0;
        for(j=0;str[j]!='\0';j++){
            if(str[i]==str[j])
                count++;
        }
        if(count==1){
            printf("First non-repeated character: %c",str[i]);
            return 0;
        }
    }
    printf("No non-repeated character found");
    return 0;
}
```

INPUT

swiss

OUTPUT

(no output)

Q2. Arrays, Array Applications and Structures

CODE

```
#include<stdio.h>
int main(){
    int n,a[100],key;
    int low,high,mid;
    scanf("%d",&n);
    for(int i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&key);
    low=0;
    high=n-1;
    while(low<=high){
        mid=(low+high)/2;
        if(a[mid]==key){
            printf("%d", mid);
            return 0;
        }
        if(a[low]<=a[mid]){
            if(key<=a[low]&&key<a[mid])
                high=mid-1;
            else
                low=mid-1;
        }
    }
    printf("-1");
    return 0;
}
```

INPUT

```
7
4 5 6 7 0 1 2
0
```

OUTPUT

```
(no output)
```

Q3. Arrays, Array Applications and Structures

CODE

```
#include<stdio.h>
int main(){
    int n,a[100][2],i,j,t,s,e;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d%d",&a[i][0],&a[i][1]);
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(a[i][0]>a[j][0]){
                t=a[i][0]; a[i][0]=a[j][0];
                t=a[i][1]; a[i][1]=a[j][1];
            }
    s=a[0][0];
    e=a[0][1];
    for(i=1;i<n;i++){
        if(a[i][0]≤e){
            if(a[i][1]>e) e=a[i][1];
        }else{
            printf("%d %d\n",s,e);
            s=a[i][0];
            e=a[i][1];
        }
    }
    printf("%d %d",s,e);
    return 0;
}
```

INPUT

```
4
1 3
2 6
8 10
15 18
```

OUTPUT

(no output)

Q4. Arrays, Array Applications and Structures

CODE

```
#include<stdio.h>
#include<string.h>
int main(){
    char str[100],temp;
    int freq[100];
    int i,j,len;
    scanf("%s",str);
    len=strlen(str);
    for(i=0;i<len;i++){
        freq[i]=1;
        if(str[i]=='\0')
            continue;
        for(j=i+1;j<len;j++){
            if(str[i]==str[j]){
                freq[i]++;
                str[j]='\0';
            }
        }
    }
    for(i=0;i<len-1;i++){
        for(j=i+1;j<len;j++){
            if(freq[i]<freq[j]){
                int t=freq[i];
                freq[i]=freq[j];
                freq[j]=t;

                temp=str[i];
                str[i]=str[j];
                str[j]=temp;
            }
        }
    }
    for(i=0;i<len;i++){
        if(str[i]!='\0'){
            for(j=0;j<freq[i];j++)
                printf("%c",str[i]);
        }
    }
    return 0;
}
```

INPUT

tree

OUTPUT

(no output)